

G1 — Disciplined Session Protocol: Recover the Golden Baseline

Adrián Lendínez Ibáñez — after the 24-Jul retrospective. Goal: re-establish the known-good state (5 clean crossings in a row, incl. 4x B→A at 52–60 s) and measure it properly before changing anything else.

THE ONE RULE THAT WAS BROKEN ON 24-JUL: one change per batch. That afternoon had 5 code edits + 3 physical changes in a single session — the collapse became unattributable. This protocol exists so that never happens again. NO code commits during a session; the only allowed git operation on the robot is checking out a validated tag.

1. Deploy the golden baseline (GPUEDGE)

```
cd ~/Documents/G1_UNITREE_ROBOT_META_REASONING
git fetch origin
git stash          # si hay cambios locales
git checkout golden-doorvis
# verificar: git log --oneline -1 -> 019ec85 "shakedown B->A con DOOR-VIS"
```

This binary = the golden window (24-Jul 16:37–16:45). It already includes: DOOR-VIS, DOOR-GUARD, DOOR-CENTER, REALCAL, marker v2 with heartbeat, per-run config snapshot. It does NOT include: RETREAT, SELF-MASK, ENV-CHANGE, OMNI (they came after — they return one at a time, twin-validated first).

2. Physical setup — frozen, and recorded in the manifest

- **Cup back IN THE HAND** (campaign-1 geometry, ARM_R=0.25): keeps the cup out of the camera frame AND restores the spill-model calibration. The chest mount broke both at once.
- **No apron** for the baseline batch. If water protection is needed later, it is its OWN experiment (one change).
- Every run command carries the physical manifest — it lands automatically in the dataset (`env_g1`):

```
G1_SETUP="cup=hand,apron=off" G1_ARM=REBASELINE ...resto del comando...
```

3. Session start ritual (unchanged)

Robot ON → app via Bluetooth → in-app teleop OK → USB → `ios_webkit_debug_proxy` → `python spill_mark.py` (ALL session — heartbeat still unvalidated on hardware!) → `noisecheck` → `reloccheck`. Perception server (320 intrinsics, NO --self-mask on the golden binary):

```
G1_FLOORCOLOR=1 python perception_server.py --host 0.0.0.0 --port 8008 --fx 300 --fy 300 --cx 160 --cy 90 --cam-h 1.10 --cam-pitch -10 --debug
```

4. Pre-registered hypothesis (write BEFORE running)

H0: with the golden binary, cup in hand, no apron, the B→A success rate is ≥70% and A→B ≥90%, times 50–110 s. If the batch does NOT reproduce this, the golden window was luck or something physical changed — and we know it with n, not vibes.

5. The re-baseline batch: 10 runs, NOTHING else

```
# DRY (no water). One line, alternating B / A. Same command all 10 runs:
G1_SETUP="cup=hand,apron=off" G1_ARM=REBASELINE G1_D00R_VIS=1 G1_META2=2 G1_M2_CFG=config_meta2_g1payload.json
G1_M2_REALCAL=1 G1_M2_STATE=meta2_state_rebase.json G1_M2_STATE_INIT=meta2_state_twin_explored.json G1_M2_L3=1 G1_M2_L4=1
G1_M2_PROFILES=1 G1_M2_D00RLIB=1 G1_M2_FRAGSPEED=1 G1_PERC=127.0.0.1:8008 python3 g1_goto.py gotoviz B
```

Rule	Why
10 runs (5 pairs), same command, no edits between runs	statistical power; any mid-batch change voids the batch
A failed run is DATA, not a bug to fix live	the fix-it-now reflex is what killed 24-Jul
Push after each pair; report weights/marks in chat	live accounting without touching the robot
If 3 consecutive runs fail the same way → STOP session, analyse offline	stopping rule agreed in advance

6. Decision gate (after the 10 runs)

- **H0 holds** → baseline re-established. Next batch: WATER at 200 g on this exact binary (cup in hand = spill model valid). Then, one at a time and twin-validated: RETREAT v2, SELF-MASK (only if apron returns), ENV-CHANGE governance.
- **H0 fails** → the difference vs 24-Jul morning is physical or environmental (lighting? furniture moved? reloc?). Diagnose from the 10 manifests — do NOT patch code.

7. What is already queued behind the gate (do not jump the queue)

Batch	Content	Status
RETREAT v2	fires only on ≥2 collisions/20 s or collision+stuck; min run age 12 s; crumb span ≥0.5 m; max 3/run; PAUSES the P9 HELP clock while reversing	coded on main, twin validation in progress

Water C2	GOV/BASE pairs 200 g, ABBA, weights via marker	waiting for baseline gate
ENV-CHANGE → governance	Renxi's parallel loop feeding meta decisions	shadow data collecting; design after C2
OMNI v3	near-field reflex fed by DEPTH scan (not laser cells)	parked opt-in